

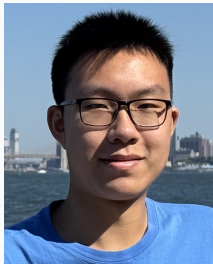
Selective Disclosure Watermarking in LLMs

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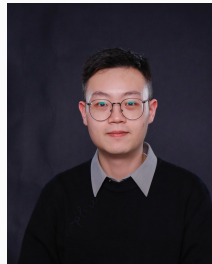
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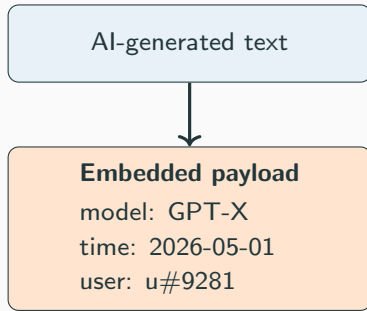
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Why Watermark LLM Outputs?

- LLM text is everywhere, so **provenance** is a real concern (EU AI Act, CA SB-942).
- **Zero-bit** watermarks only tell us “AI or human”. They cannot say which model, when, or for which user.
- **Multi-bit** watermarks embed metadata such as model id, timestamp, and user or session id.

The Problem

Existing multi-bit schemes are **all-or-nothing**: anyone with the verification key reads the entire payload.



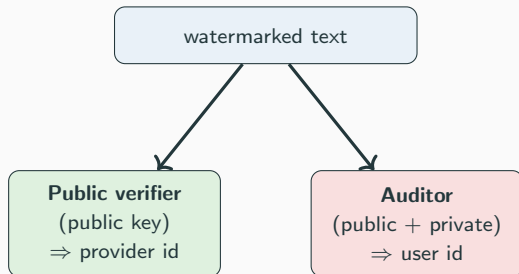
Different Verifiers Need Different Information

Example (one of many possible settings):

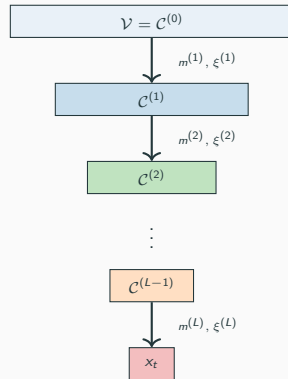
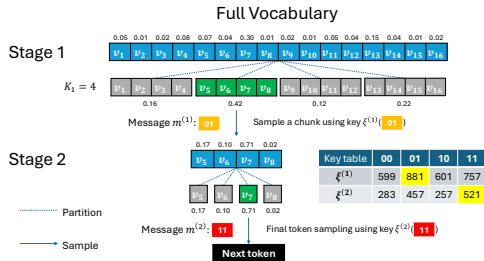
- **Public verifier** (e.g., a platform) only needs to know “is this from provider X?”
- **Privileged auditor** needs the user or session id, under due process.

Our Goal: Selective Disclosure

Each verifier decodes **only the payload portion** they are authorized to see, and nothing else.



Key Idea: Hierarchical Vocabulary Routing (HeRo)



- Partition the vocabulary into chunks; route through a **nested tree**.
- Level ℓ uses its **own key** $\xi^{(\ell)}$ to carry payload segment $m^{(\ell)}$.
- The selected chunk becomes the vocabulary for the next level, and we repeat the same step.

Theorem 1 (Statistical Unbiasedness)

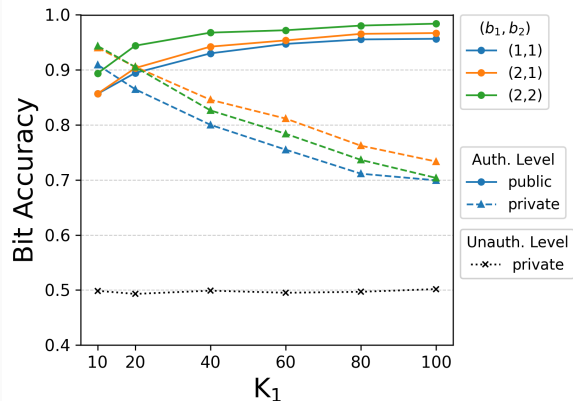
The final sampled token x_t follows the original next-token distribution P_t .

⇒ **Text quality is preserved.**

Theorem 2 (Selective Disclosure)

A verifier holding only $\xi^{(1)}, \dots, \xi^{(k)}$ cannot recover $m^{(\ell)}$ for any $\ell > k$ beyond random guessing.

Experiment 1: Selective Disclosure Works

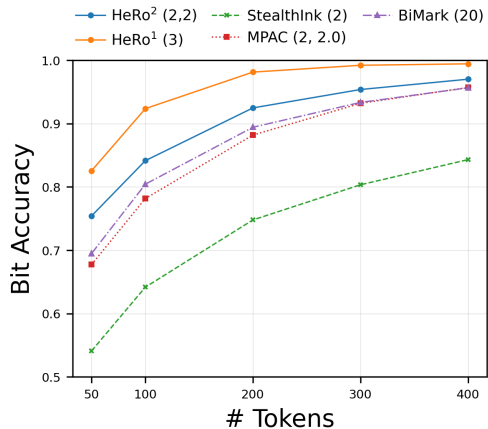


Two-level payload, Llama2-7B on C4 realnewslike.

- **Public** accuracy increases with K_1 (more public signal).
- **Private (authorized)** stays decodable across the curve.
- **Private (public-only key)** $\approx 50\%$: **random guess**, as Theorem 2 predicts.

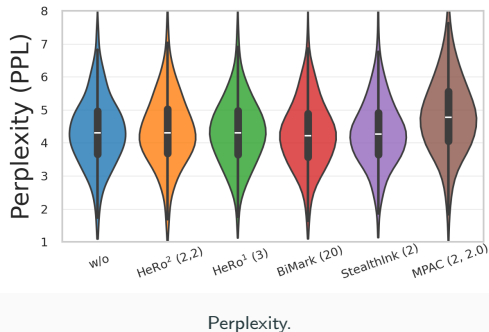
Sweet spot at $K_1 = 20$.

Experiment 2: Detectability and Text Quality

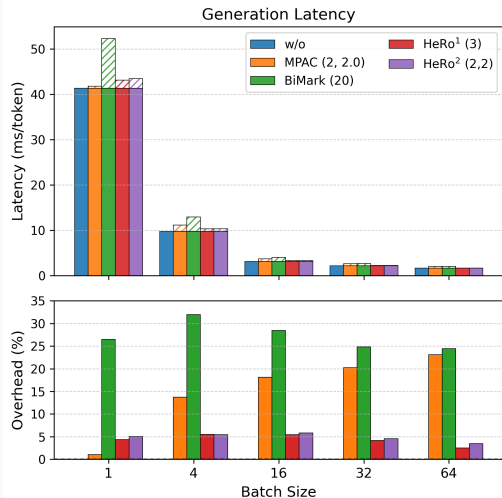


Bit accuracy vs. token budget.

- **Higher accuracy with fewer tokens** than MPAC, StealthInk, BiMark.
- PPL distribution closely overlaps with the unwatermarked text, consistent with unbiasedness guarantee (Theorem 1).



Experiment 3: Efficiency



Generation latency (top: ms/token; bottom: % overhead).

Decoding latency (μ s/token)

Method	Batch Size				
	1	4	16	32	64
MPAC	488	479	491	489	480
BiMark	130	130	131	132	133
HeRo ¹	7.8	2.4	1.7	1.6	1.4
HeRo ²	10.3	3.1	1.9	1.6	1.5

- Publicly released codebases: MPAC, BiMark.
- Generation overhead only $\sim 5\%$** vs. MPAC and BiMark ($>20\%$).
- Decoding $\sim 10^2\times$ faster than MPAC and BiMark.**

HeRo

Hierarchical Vocabulary Routing for Selective Disclosure Watermarking in LLMs

Multi-bit watermarking with **multi-level access control**.

Code <https://github.com/xuyangc03/hero-watermark>

Thank you!

